SBS MODIFIED BITUMEN WATERPROOFING MEMBRANE

INSTALLATION MANUAL

Torch-applied reinforced sheet system

SBS ELASTOMER-MODIFIED BITUMEN MEMBRANE

ROOFING | BASEMENT | PODIUM | CIVIL WORKS

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IMPORTANT

General guidance for trained professional applicators. Approved design, local codes, project specifications and the product-specific method statement take precedence.

1. PURPOSE, SCOPE AND SYSTEM RESPONSIBILITY

This manual describes general field procedures for installing Tianyuan SBS modified bitumen waterproofing membranes by controlled torch application. It covers common horizontal and vertical waterproofing work on roofs, basements, podiums and selected civil structures.

Suitable scope

Reinforced SBS membranes with a thermofusible lower film, used as a base ply, intermediate ply or cap sheet within an approved system.

Project design

The designer must define substrate, drainage, number of plies, membrane thickness, protection, insulation, movement joints and detail geometry.

Installer responsibility

The installer accepts the substrate, confirms compatibility, controls hot work, records inspections and corrects defects before covering.

System compatibility

Primer, base sheet, insulation, cover board, protection board, sealant and accessory materials must be mutually compatible and approved for the assembly.

1.1 Typical system build-up

Layer	Component	Function
1	Protection / finish	Mineral-surfaced cap sheet, screed, protection board or project finish
2	Waterproofing membrane	One or more SBS modified bitumen membrane layers
3	Primer / receiving layer	Dry bitumen primer, compatible base sheet or approved cover board
4	Substrate	Concrete, masonry, metal, approved board or prepared existing bituminous surface
5	Structural support	Roof deck, wall, foundation slab or civil structure

SELECTION NOTE SBS membranes are generally selected where flexibility, fatigue resistance and low-temperature performance are important. Exposed roofing and permanently submerged work require specific product and system approval.

1.2 Conditions requiring technical approval

- Direct torching to combustible substrates, insulation or concealed cavities.
- Application over damp concrete, fresh concrete, uncured coatings, solvent-rich adhesive or contaminated surfaces.
- Below-grade work subject to hydrostatic pressure, potable-water contact, chemical exposure or negative-side water pressure.
- Cold-weather work, steep slopes, high wind zones, large movements or unusual penetrations.
- Any detail that differs from the approved drawings or local waterproofing code.

2. MATERIALS, TOOLS AND SITE PREPARATION

2.1 Required materials

SBS membrane Correct thickness, reinforcement, surface finish and roll length. Check labels and batch identification.	**Bitumen primer** Compatible primer for porous or specified substrates. Use only after confirming drying time and coverage.
Detail materials Reinforcing strips, compatible membrane patches, termination bars, metal flashings and approved sealants.	**Protection materials** Protection board, drainage board, insulation, separation layer or screed as defined by the design.

2.2 Typical tools and equipment

Work stage	Typical equipment
Layout and cutting	Tape measure, chalk line, straight edge, roofing knife, spare blades, shears
Surface preparation	Broom, industrial vacuum, scraper, grinder with dust control, repair tools
Primer application	Roller, brush or approved spray equipment; clean mixing and transfer containers
Torch application	Approved propane torch, hose, regulator, cylinder trolley, torch stand and leak-test solution
Seaming and detailing	Weighted roller, hand roller, rounded trowel, heated trowel and granule embedment tools
Inspection	Seam probe, wet-film or coverage checks where applicable, thermometer, moisture meter and camera

2.3 Personal protective equipment

- Flame-resistant work clothing appropriate to the hot-work risk assessment.
- Heat-resistant gloves, safety footwear, eye protection and hard hat.
- Fall protection, edge protection and safe access equipment where required.
- Respiratory and hearing protection where the site exposure assessment requires them.

DELIVERY AND STORAGE Keep rolls upright on a flat, dry surface, protected from weather, heat, sparks and physical damage. Do not use wet, distorted, crushed or unlabeled rolls. Condition cold rolls before unrolling to reduce cracking and wrinkling.

3. HOT-WORK, FIRE AND OCCUPATIONAL SAFETY

HIGH-RISK ACTIVITY Torch application creates open-flame, burn, fire, smoldering-fire and fume hazards. A written hot-work plan, trained personnel and continuous supervision are required.

3.1 Before ignition

1. Inspect the roof or structure above and below the work area. Identify combustible decks, insulation, timber blocking, voids, vents, gas lines, cables and stored materials.
2. Obtain the required hot-work permit and coordinate shutdown or protection of nearby air intakes and sensitive equipment.
3. Remove combustible debris, packaging, solvent containers and loose waste from the torching zone.
4. Provide serviceable fire extinguishers immediately available to the torch crew. Follow local requirements for extinguisher type, size and spacing.
5. Leak-test hoses, regulators, valves and connections with an approved solution. Never use a flame for leak testing.
6. Confirm safe cylinder location, upright restraint, hose routing and emergency shutoff access.

3.2 During application

- Keep the flame moving. Do not direct flame into gaps, penetrations, cavities, under flashings or toward combustible components.
- Never leave an ignited torch unattended. Place it only on an approved stand and extinguish it when not actively heating.
- Maintain clear communication between torch operator, roll handler and fire-watch personnel.
- Stop work when wind, visibility, access, fumes, weather or site activity prevents safe control of the flame.
- Use flame-free detailing methods where open flame cannot be safely controlled.

3.3 Fire watch and closeout

- Provide exterior and interior fire watch where heat could transmit through the assembly or enter concealed spaces.
- Inspect the work area, adjacent areas and the underside of the deck throughout application and for the duration required by the hot-work permit and local rules.
- Remove cylinders and ignition sources from the work zone at the end of the shift. Record the final fire-watch inspection.

HEALTH CONTROL Avoid unnecessary smoke and overheating. Provide ventilation and apply exposure controls identified by the site risk assessment. Wash exposed skin and follow the product Safety Data Sheet.

4. SUBSTRATE ACCEPTANCE AND PREPARATION

4.1 Acceptance criteria

Item	Acceptance requirement
Condition	Structurally sound, stable and able to support the specified waterproofing system and construction traffic.
Surface	Smooth, even and free of sharp projections, fins, laitance, loose aggregate, dust and debris.
Moisture	Dry enough for the specified primer and membrane system. No standing water, frost or visible dampness.
Contamination	Free of oil, grease, curing compounds, release agents, incompatible coatings and solvent residue.
Falls and drainage	Positive falls, correctly set drains and no unapproved depressions that will trap water.
Joints and cracks	Movement joints, construction joints and cracks treated in accordance with approved details.
Corners	Internal corners provided with approved fillets or cant strips where required; external corners eased and reinforced.
Openings	Penetrations, outlets, sleeves and embeds secure and ready for waterproofing.

4.2 Preparation sequence

1. Complete structural repairs and allow repair materials to cure.
2. Grind or remove projections and fill voids, honeycombs and open joints with compatible repair material.
3. Clean by brooming and vacuuming. Use controlled mechanical preparation where necessary.
4. Confirm substrate moisture with the project-approved test method.
5. Protect adjacent finishes, identify detail zones and record substrate acceptance before priming.

CONCRETE NOTE New concrete requires adequate curing and moisture reduction. A calendar age alone does not prove suitability. Verify dryness and primer compatibility before membrane application.

4.3 Primer application

- Apply a thin, uniform coat at the product-approved coverage rate.
- Avoid puddles, runs and heavy build-up. Keep primer out of joints that require movement treatment.
- Allow primer to dry completely. Do not torch over wet or tacky solvent-rich primer.
- Re-prime surfaces that become contaminated, weathered or remain exposed beyond the primer manufacturer's permitted interval.

5. LAYOUT AND TORCH APPLICATION

Figure 1. Original installation workflow prepared for this manual.

5.1 Roll layout

- Start at the lowest point and arrange courses so laps shed water toward outlets.
- Dry-lay or partially unroll each sheet, align to control lines and check planned lap widths.
- Allow rolls to relax as site temperature and product condition require. Re-roll evenly from one or both ends without disturbing alignment.
- Offset side and end laps between successive plies. Avoid four-layer build-up at intersections.
- Cut manageable lengths for vertical work, flashings and complex details.

5.2 Torch-welding sequence

1. Position the roll and confirm the receiving surface is clean and dry.
2. Ignite and adjust the torch away from combustible materials. Use a stable, controlled flame.
3. Heat the membrane underside and receiving surface with continuous sweeping movement. The release film should melt completely where present.
4. Advance the roll slowly while maintaining a small, continuous flow of softened bitumen at the leading edge.
5. Press and roll the sheet into full contact. Do not trap air or walk on unsupported hot membrane.
6. Repeat from the opposite end where the sheet was re-rolled. Keep the sheet flat and aligned.
7. Immediately inspect the installed area for wrinkles, voids, dry spots, overheating and displaced reinforcement.

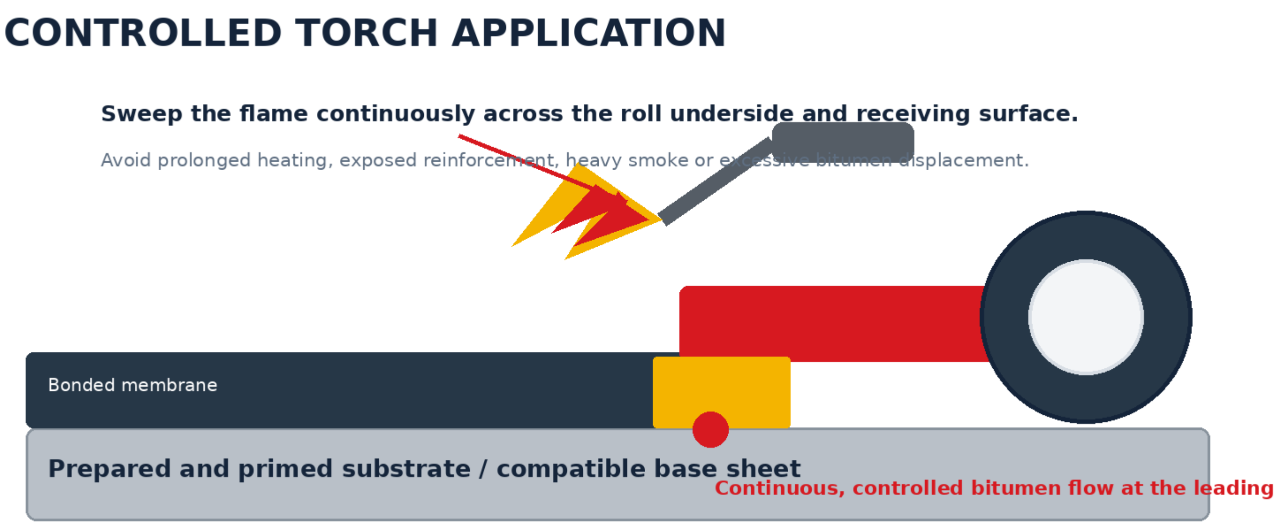

Figure 2. Controlled heating concept. The diagram is schematic and does not replace applicator training.

6. LAPS, SEAMS AND FIELD JOINTS

TYPICAL LAP GEOMETRY

Use the approved project detail where it requires wider laps. Stagger adjacent end laps and orient laps to shed water.

Figure 3. Typical lap geometry. Use project-approved dimensions where they are greater.

6.1 Typical lap requirements

Joint	Typical guide	Execution requirement
Side lap	80 mm minimum	Use the factory selvage where provided. Maintain straight, fully bonded seams.
End lap	100-150 mm typical	Use the approved system value. Stagger adjacent end laps by at least 300 mm where practical.
Granulated surface lap	Prepare a bondable bitumen surface	Embed or remove loose granules without damaging the reinforcement, then heat and consolidate the seam.
Vertical flashing lap	80 mm minimum unless detailed otherwise	Use wider laps where the seam is formed over granules or at high-risk transitions.
Patch overlap	100 mm minimum beyond sound material	Round patch corners and fully bond the perimeter.

6.2 Seam formation and inspection

- Heat both mating surfaces uniformly and produce a visible, continuous bitumen bead without excessive flow.
- Use a hand roller or weighted roller while the seam remains workable. Apply pressure from the closed side toward the exposed edge.
- Probe cooled seams using a rounded seam probe. Mark, open and repair every unbonded location.
- On mineral-surfaced sheets, broadcast matching granules into the warm bitumen bead where appearance requires it.
- Do not conceal seams until the inspection and repair record is complete.

DO NOT ACCEPT Dry laps, open edges, fishmouths, bridging, wrinkles, scorched compound, exposed or distorted reinforcement, channels through the seam, or laps facing against water flow.

7. FLASHINGS AND CRITICAL DETAILS

CRITICAL DETAIL PRINCIPLES

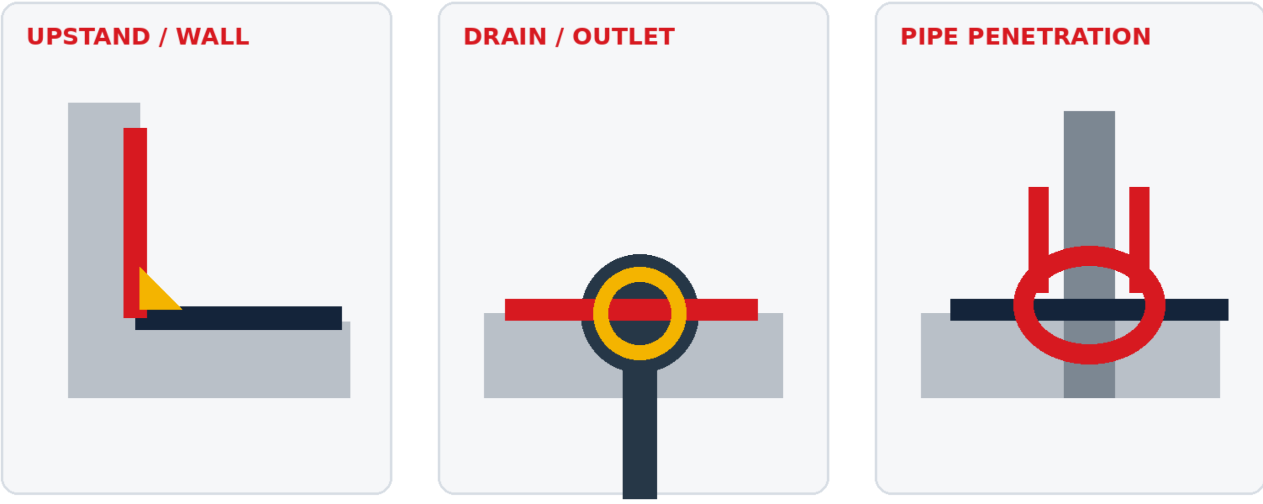

Figure 4. General detail principles. Final dimensions and components must follow approved drawings.

7.1 Internal and external corners

- Reinforce corners before installing the main flashing layer.
- Avoid stretching the membrane around a corner. Cut and fit pieces so the sheet remains fully supported.
- Round patch corners and stagger cut lines between reinforcement layers.

7.2 Upstands and terminations

- Carry the flashing to the design height above the finished waterproofed surface and expected water level.
- Fully bond the membrane to the vertical substrate without bridging at the base of the upstand.
- Mechanically secure the upper edge using the approved termination bar, counterflashing or chase detail.
- Seal the termination with a compatible sealant and protect it from direct water entry.

7.3 Drains and outlets

- Provide a stable recessed sump with positive fall to the outlet.
- Install additional reinforcement around the outlet and fully bond the membrane into the flange or clamping assembly.
- Keep the drainage opening clear. Inspect for wrinkles and channels that could bypass the clamping zone.

7.4 Penetrations and movement joints

- Use compatible sleeves, collars and preformed components where available.
- Secure and seal the top of pipe flashings. Maintain separation from hot pipes where the membrane temperature limit could be exceeded.
- Do not bridge structural movement joints with fully bonded field membrane. Install the specified movement-joint assembly.

> **DETAIL CONTROL** Details are the highest-risk areas of the waterproofing system. Prepare a sample detail or trial area when the project configuration is unfamiliar.

8. INSPECTION, REPAIRS AND PROTECTION

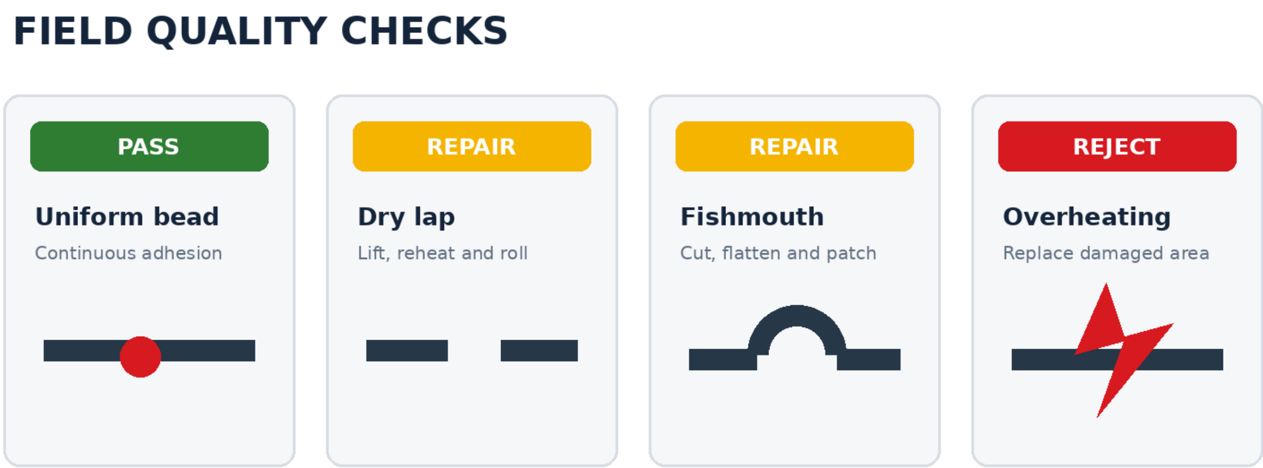

Figure 5. Common field quality checks and corrective actions.

8.1 Inspection stages

Stage	Required checks
Before work	Substrate acceptance, weather, permits, materials, equipment, detail readiness and protection.
During installation	Alignment, full bonding, controlled heating, lap width, continuous bead, wrinkle control and detail sequencing.
Daily closeout	Open edges temporarily sealed, drains clear, tools removed, fire watch completed and work protected.
Before covering	All seams probed, repairs completed, photographs and batch records filed, protection method approved.
Final handover	Punch list closed, drawings and inspection records submitted, maintenance limitations explained.

8.2 Repair methods

- Dry or open lap: gently lift the edge, clean, reheat both faces, press and roll until a continuous bond is achieved.
- Fishmouth or wrinkle: cut the raised area to release stress, heat and flatten, then install a rounded patch over sound bonded membrane.
- Puncture or cut: remove loose material, prepare the surface and install a fully bonded patch extending at least 100 mm beyond the damage.
- Overheated or exposed reinforcement: cut out the affected membrane to sound material and install a reinforced replacement patch.
- Blister: determine and correct the moisture or adhesion cause before cutting, flattening and patching.

8.3 Protection after installation

- Limit traffic and use designated walkways.
- Install protection boards, drainage layers, insulation or screeds without dragging sharp materials across the membrane.
- Use compatible separation layers under concrete, pavers or backfill where specified.
- Prevent reinforcement steel, pallets, equipment legs and follow-on trades from puncturing the membrane.
- For below-grade work, inspect immediately before backfilling and place protection progressively.

9. WEATHER, TEMPERATURE AND SPECIAL CONDITIONS

9.1 Weather limitations

- Do not install during rain, snow, frost, heavy condensation or when water can enter the work area.
- Stop when wind prevents stable roll control, safe torch operation or uniform heating.
- Cold rolls can crack, curl or resist bonding. Store and condition materials in accordance with the project method statement.
- In cool conditions, increase preparation and seam inspection. Do not compensate by scorching the membrane.
- Protect hot membrane from sudden rain and from construction traffic until it has cooled and stabilized.

9.2 Common problems and corrective control

Problem	Likely cause	Corrective control
Membrane will not lie flat	Cold roll, poor storage or insufficient relaxation	Condition the roll, allow relaxation and replace damaged material.
Dry seam	Insufficient or uneven heat, contaminated lap	Open, clean, reheat and roll. Increase inspection frequency.
Heavy bitumen flow	Excess heat or slow advance	Reduce heat exposure, maintain steady movement and replace damaged areas.
Wrinkles / fishmouths	Misalignment, trapped air or roll tension	Stop, realign and remove stress before continuing. Patch as required.
Blistering	Moisture, vapor pressure or incomplete bonding	Identify the source. Do not patch without correcting the cause.
Poor adhesion to substrate	Dust, dampness, incompatible or wet primer	Remove affected area and prepare the substrate correctly.
Open lap over granules	Inadequate granule preparation	Create a clean bitumen bonding zone, reheat and consolidate.
Damage by following trades	Insufficient access control or protection	Repair, document and install protective layers before work resumes.

SPECIAL EXPOSURE Chemical contact, hydrostatic pressure, green roofs, potable water, vehicle decks, high-temperature surfaces and root resistance require a specifically approved membrane and system design.

10. SITE CHECKLISTS

10.1 Pre-start checklist

No.	Check	Requirement
1	[ ]	Approved drawings, method statement and product TDS available on site.
2	[ ]	Applicators trained; supervisor and fire-watch personnel assigned.
3	[ ]	Hot-work permit issued and emergency arrangements confirmed.
4	[ ]	Substrate accepted and documented; moisture condition acceptable.
5	[ ]	Falls, drains, penetrations, joints and terminations ready.
6	[ ]	Membranes, primers and accessories verified for type, batch and condition.
7	[ ]	Torch, hoses, regulators and cylinders inspected and leak-tested.
8	[ ]	Fire extinguishers, PPE, fall protection and access controls in place.
9	[ ]	Weather suitable and forecast reviewed.
10	[ ]	Trial area or first-off detail approved where required.

10.2 Final inspection checklist

No.	Check	Requirement
1	[ ]	All field sheets fully bonded and aligned.
2	[ ]	Side laps and end laps meet approved dimensions.
3	[ ]	Seams cooled, probed and repaired; no open edges.
4	[ ]	Corners, upstands, drains and penetrations complete.
5	[ ]	Termination bars, flashings and sealants installed.
6	[ ]	No wrinkles, fishmouths, punctures, blisters or exposed reinforcement.
7	[ ]	Drainage paths clear and outlets functional.
8	[ ]	Protection layer installed or membrane isolated from follow-on work.
9	[ ]	Fire watch completed and documented.
10	[ ]	Photographs, batch records, inspection sheets and repair log filed.

RECORDS Retain delivery tickets, batch labels, substrate acceptance, weather log, permit, inspection photographs, repair locations and sign-off documents as part of the project quality record.

11. TECHNICAL BASIS AND DOCUMENT CONTROL

This manual is an independently written Tianyuan Waterproof document. The text, tables and diagrams were created specifically for this publication. No competitor manual text, layout or illustrations have been reproduced.

11.1 Public technical references consulted

Organization	Reference basis
Polyglass U.S.A.	Guide Spec No. 200 SBS Torch, SBS cap sheet plus base sheet system, February 2021.
Polyglass U.S.A.	Technical Guide for modified bitumen roofing systems, 2023 edition.
GAF	RUBEROID APP Torch-Applied Roofing Systems Manual, Version 1.0, used for general torch safety, seam control and hot-work principles.
Occupational Safety and Health Administration	Fire-prevention requirements for heating and hot work, and asphalt-fume control resources.
ASTM International	ASTM D41 for asphalt primer and ASTM D6164 for SBS modified bituminous sheet materials using polyester reinforcement.
Project authority	Applicable national and local building, fire, occupational safety and waterproofing requirements.

COPYRIGHT POSITION Public technical sources were used to identify established industry principles. All instructions have been summarized, reorganized and expressed in original language. Product-specific dimensions, warranties and proprietary system details from other manufacturers are excluded.

11.2 Limitations

- This document does not design the waterproofing system or replace engineering review.
- The project specification and approved detail drawings govern where they differ from this manual.
- Field conditions vary. The installer must stop and obtain technical direction when safe, durable installation cannot be assured.
- Product data and installation requirements may be revised. Confirm the current approved revision before use.

11.3 Document identification

Field	Value
Document title	SBS Modified Bitumen Waterproofing Membrane Installation Manual
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TIANYUAN WATERPROOF

Technical documentation for professional waterproofing systems